HENRY WILLIAMS / 911RECORDS.ORG

From scanned records to cited answers.

An independent research platform for New York City's released September 11 records.

64,385	PDFs available	**231,104**	catalog pages	**Nightly**	catalog refresh + change tracking

01 / BUILD THE EVIDENCE LAYER

INGESTION
City catalog + PDFs
Snapshot and diff; track changes and withdrawals. Preserve source files and Bates page IDs.

OCR / LOCAL
Extract text + recover scans
Embedded text > Tesseract fallback > 300-DPI Apple Vision / Tesseract comparison. Readability, cross-engine agreement and number-preservation gates control replacements.

PROVENANCE
Selected page text
Use the current catalog copy and approved OCR. Keep each page linked to its original scan.

NLP / LANGUAGE PROCESSING
Structured information
Python regex + dictionaries extract addresses, labs, dates and measurements. Fuzzy matching resolves names; rules classify record types.

ML
Meaning + discovery
Local nomic-embed-text: 768-D vectors from page chunks. Similarity search, related records and scikit-learn topic clustering.

LLMs
Readable descriptions
Local Qwen: titles and summaries. Claude Haiku: topic labels, fallback facts and entity cleanup.

PUBLISH PostgreSQL: metadata + text | OpenSearch: text + vectors | Next.js: two hosts behind Cloudflare

serves retrieval

02 / SEARCH, EXPLAIN, VERIFY

INPUT
Researcher asks
Bates ID: open the record.
Keywords: search directly.
Question: use an LLM plan.

ROUTING
Plan + retrieve
Haiku plans question terms and filters. Embed the query locally using the same nomic model.

HYBRID SEARCH
Rank the evidence
Filter eligible pages; rank with OpenSearch BM25 + k-NN. Normalize and blend: 40% keyword / 60% semantic.

USER EXPERIENCE
Results you can check
Search: snippets + filters. Ask: Haiku uses up to 12 excerpts with checked citation IDs. Verify in the scan + OCR viewer.

EXPERIENCE DEMONSTRATED

ML Embedding pipelines, clustering and OCR quality evaluation.

NLP Information extraction, canonicalization and document classification.

LLMs Local inference, structured outputs and retrieval-augmented generation.

Pretrained models integrated into a production system. Original scans remain the evidence.

Corpus: October 27, 2026